CV

- Camera Model
  - Pinhole Camera Model
    - $x = \frac{z+f}{f}x_i$
    - $y = \frac{z+f}{f}y_i$
  - Homogeneous Coordinates
    - Definition
      - Add a scale factor as the new dimension: $(x, y, z, 1) \sim (kx, ky, kz, k), \quad \text{for any } k \neq 0$
      - k=0 represent the plane at infinity
      - Allow various image transformations to be easily represented by a matrix
    - For pinhole camera model
      - $x_i = (x, y, z, k)$
      - $x = (x, y, z, 1)$
      - $k = \frac{z+f}{f}$
  - f>>d: Orthographic projection
    - $x = x_i, y = y_i$
    - Objects remain the same size regardless of distance
    - Projection rays are parallel
    - Otherwise: perspective projection
- Image Geometry
  - **Degrees of Freedom (DoF)**: the number of independent parameters that define the transformation
  - Translation $\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$, DoF=2
  - Rotation $\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) & t_x \\ -\sin(\alpha) & \cos(\alpha) & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$ For $\theta > 0$, rotation is **closewise**, DoF=3
  - Similarity $\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s\cos(\alpha) & s\sin(\alpha) & t_x \\ -s\sin(\alpha) & s\cos(\alpha) & t_y \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$, DoF=4
  - Affine $\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x\cos(\alpha) & s_x\sin(\alpha) & t_x \\ -s_y\sin(\alpha) & s_y\cos(\alpha) & t_y \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$, DoF=6
    - $x'' = \frac{a_{11}x+a_{12}y+a_{13}}{0\cdot x+0\cdot y+1}$
    - $y'' = \frac{a_{21}x+a_{22}y+a_{23}}{0\cdot x+0\cdot y+1}$
  - Projective $\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$, DoF=8
    - $x'' = \frac{h_{11}x+h_{12}y+h_{13}}{h_{31}x+h_{32}y+1}$
    - $y'' = \frac{h_{21}x+h_{22}y+h_{23}}{h_{31}x+h_{32}y+1}$
  - Approximation of Projective Transformation by Affine Transformation
    - $h_{31}, h_{32} \approx 0 \Rightarrow h_{31}x + h_{32}y + 1 \approx 1$ (Far-field scenario, negligible perspective effects)
    - $x, y \approx 0 \Rightarrow h_{31}x + h_{32}y + 1 \approx 1$ (Small-scale region, minimal perspective distortion)
- Image Matching
  - Image Matching
    - Keypoint detection
    - Feature description
    - Similarity Measure
  - Transformation Estimation
    - Homography Matrix $x_L = Hx_R$
    - Fundamental Matrix $x_L^{\mathrm{T}}Fx_R = 0$
      - Epipole
      - Epipolar Line
  - Camera Callibration
    - Pinhole Camera Model
      - Normalized Image Plane $\hat{p} = [\hat{x}, \hat{y}, 1]^T = \frac{1}{z}\left[\mathcal{I}, 0\right] p = \left[x/z, y/z, 1\right]^T$
      - Camera Projection $p = \mathcal{M}P = K[R|t]P$
        - Intrinsic Parameters K
          - Ideal $K = \begin{bmatrix} -fs_x & 0 & x_0 \\ 0 & -fs_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}$
            - focal length f
            - principal point $c_0 = (x_0, y_0)$
            - pixel size (s_x, s_y)
          - General $K = \begin{bmatrix} \alpha & -\alpha\cot\theta & x_0 \\ 0 & \frac{\beta}{\sin\theta} & y_0 \\ 0 & 0 & 1 \end{bmatrix}$
        - Extrinsic Parameters
          - R rotation matrix
          - t translation vector
        - $\mathcal{M} = \begin{bmatrix} \alpha r_1^{\mathrm{T}} - \alpha\cot\theta r_2^{\mathrm{T}} + x_0 r_3^{\mathrm{T}} & \alpha t_x - \alpha\cot\theta t_y + x_0 t_z \\ \frac{\beta}{\sin\theta} r_2^{\mathrm{T}} + y_0 r_3^{\mathrm{T}} & \frac{\beta}{\sin\theta} t_y + y_0 t_z \\ r_3^{\mathrm{T}} & t_z \end{bmatrix}$
    - Camera Callibration $P' = \mathcal{M}P_w \Rightarrow \begin{bmatrix} x_i' \\ y_i' \\ z_i' \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$
    - Homogeneous Linear System $\mathcal{P}\left[\, m_{11}, m_{12}, m_{13}, m_{14}, m_{21}, m_{22}, m_{23}, m_{24}, m_{31}, m_{32}, m_{33}, m_{34} \,\right]^T = 0$
  - Camera Distortions
    - Chromatic Aberration
    - Spherical Aberration
    - Radial Distortion
      - Barrel Distortion
      - Pincushion Distortion
      - Mustache Distortion
- Stereo Vision
  - Stereo Setup
    - obtain depth information from disparity $d = x_L - x_R$
    - Simple stereo model: with parallel optical axes
      - parameters
        - focal length f
        - baseline b
        - estimated depth $Z = \frac{f\cdot b}{d}$
        - depth is inversely proportional to disparsity
      - problems
        - prior knowledge of parameters
        - correspondence
  - Epipolar Geometry
    - Epipolar plane
      - connects **C1, C2, P**
      - cuts through image planes forming epipolar lines
      - Corresponding **P1, P2** must lie on their epipolar lines (reduced search space)
    - Epipolar lines all intersects at epipoles **e**
    - Can be obtained from **fundamental matrix** (relative orientations of cameras)
  - Stereo Rectification
    - Definition
      - Reproject image planes onto a common plane parallel to the line between the optical centres
      - Make epipolar lines horizontal (corresponding points will be horizontal)
      - Make epipoles at infinity
      - Simplify the model into parallel stereo model
    - Steps
      - Estimate $[R|t]$ from fundamental matrix
      - Estimate the epipole e (SVD of $[R|t]$)
      - Build R_{rect} from e (to make epipoles at infinity)
      - Decompose $[R|t]$ into R and t
      - Set $R_1 = R_{rect}$ (left camera) and $R_2 = RR_{rect}$ (right camera)
      - Apply rotation and sacalation to both images
  - Stereo Correspondence
    - Algorithm
      - For each epipolar line
      - - For each pixel (window) in the left image
      - -- compare with each pixel (window) on the same line in the right image
      - -- pick pixel (window) with minimum match cost
    - Properties
      - Match quality
      - Smoothness (adjacent / similar disparities)
    - Stereo disparity map
    - Effect of window size
      - Smaller: good precision, more detailed; sensitive to noise
      - Larger: robust to noise; reduced precision, less detailed
  - Stereo Reconstruction
    - light projector + camera
    - establish a virtual image plane for the illuminant
    - calibrate the light projector and the camera
- Model fitting
  - Hough Transform
    - Definition
      - Each candidate point votes for all lines it is consistent with
      - Representation of a line
        - Usual form $ax + by = d$
        - Improved form $x\cos\theta + y\sin\theta = \rho$
      - The brightest point in the transformed image is the best parameters
    - Steps
      - Define model and its parameters
      - Derive a formula to obtain parameter values given a data point
      - Define quantisation steps and limits of the parameters
      - Build accumulative array for the parameter space (Hough space)
      - For each data point
        - Compute parameter values for a data point and increment the cell in the array that corresponds to these values
      - Perform non maxima suppression in the array
      - Recover the parameter values that correspond to the maxima
  - Generalised Hough Transform
    - Localising free form shapes / features by voting
    - 1. Constructing the shape model
      - Select a reference point
      - Construct a table including $\phi, (r, \alpha)$
    - 2. Find location of the shape in the image (voting)
    - Advantages
      - Can extract feature groupings from clutter in linear time
      - Match objects in noisy and complex environments
    - Problems
      - models with M parameters require M-dimensional accumulator array
        (solution: count votes for each model without the accumulator array)
      - outliers (sample data to find a set of inliers)
  - Least Squares
    - Basics
      - For line $ax + by = d$, find best a, b, d to match the dataset
      - Minimise $\mathcal{L}(a, b) = \sum_{i=1}^{n}(ax_i + by_i - d)^2$
      - Sensitive to outliers
    - Improvement
      - Use a better error $\rho(u, \sigma) = \frac{u^2}{\sigma^2+u^2}$ to penalise outliers
        - behaves like squared distance for small values of the residual u
        - saturates for larger values of u
      - Scale σ needs to be well chosen
  - RANSAC - Random Sample Concensus
    - robust fitting of models in the presence of many data outliers
    - Steps
      - Select s data points at random to form a sample
        (minimal set – the smallest number of samples from which the model can be computed)
      - Estimate model parameters from s data points
      - Find how many inliers C (points within distance/error t) are in the data (concensus set C_k)
      - Maximise $|C_k|$
    - Notes
      - The extent of fitting can be adjusted by t / the proportion of the whole set
      - If there are multiple structures then, after a successful fit, remove the fit data and redo RANSAC
      - t is set empirically according to noise variance (e.g. $t^2 = 3.84\sigma^2$)
      - Number of trails: $N > \frac{\log(1-p)}{\log(1-(1-e)^s)}$
        - needed for at least one correct set (inliers only), with probability p
        - N grows quickly with e and s
        - Inlier ratio e is often unknown a priori, so pick worst case, e.g. $N = \infty$, and adapt if more inliers are found